

RadiX Dose

v2.3

Build an example clinic & run your first calculation.

This guide walks you through RadiX Dose from an empty install: first you'll **build a small example clinic** (a facility, a machine, a chamber and a beam), then you'll **run a reference-dosimetry calculation** with it. The example data used here is entirely fictional — use it to learn the workflow, then replace it with your own.

A note on the data

"General Hospital / Radiation Oncology" is a neutral example only. Never store patient-identifiable information in the app — it is built for quality-assurance data.

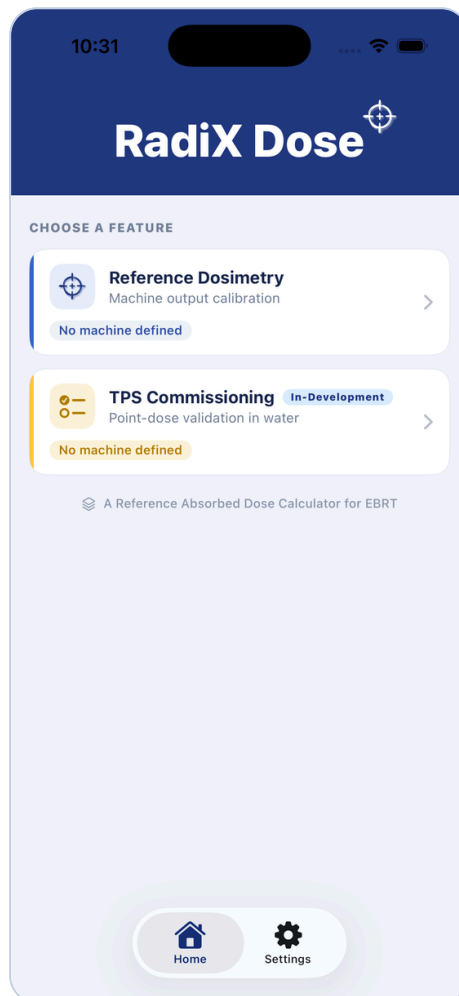
01 Open the app

1 Launch RadiX Dose and accept the agreement.

On first launch the **User Agreement & Disclaimer** appears. Read it, tick the box, and tap **Accept** (this is asked once).

2 Choose a feature.

On the home screen, tap **Reference Dosimetry** ("Machine output calibration"). The second feature, *TPS Commissioning*, is a preview and not needed here.



The home screen — tap Reference Dosimetry to begin.

02 Build the example clinic

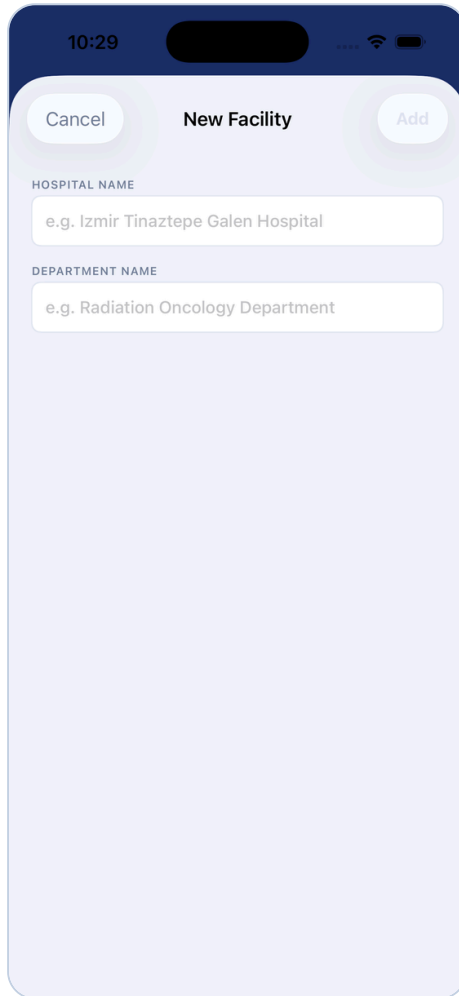
Everything a calculation needs is defined once, in the **Settings** tab (bottom-right), as a nested library: **Facility** → **Machine** → **Detectors** · **Electrometers** · **Beams**. One facility and one machine are *active* at a time.

1 Add a facility.

Go to **Settings**. Under **FACILITY**, tap **+ Add**, then fill in:

- **Hospital Name** — e.g. General Hospital
- **Department Name** — e.g. Radiation Oncology

Tap **Add**. The facility appears and becomes **ACTIVE**.

A screenshot of a mobile application interface for adding a new facility. The screen is titled "New Facility" and has a light purple background. At the top, there is a status bar showing the time "10:29" and battery level. Below the title bar, there are three buttons: "Cancel" on the left, "New Facility" in the center, and "Add" on the right. The "Add" button is highlighted with a green glow. Below the buttons, there are two text input fields. The first field is labeled "HOSPITAL NAME" and contains the placeholder text "e.g. Izmir Tinaztepe Galen Hospital". The second field is labeled "DEPARTMENT NAME" and contains the placeholder text "e.g. Radiation Oncology Department".

New Facility — hospital and department names.

2

Add a machine (linac).

Tap the **chevron** to the left of your facility to expand it, then tap **+ Add Machine** and set:

- **Manufacturer & Model** — e.g. Varian · TrueBeam
- **Machine ID / Serial** — e.g. LINAC-1
- **Reference Absorbed Dose Protocol** — IAEA TRS-398 Rev.1 (2024), IAEA TRS-398 (2006), or AAPM TG-51. This chooses which beam-quality index and k_Q data the app uses.
- **Setup Configuration** — SSD- or SAD-based
- **Dose Reporting Point** — report at d_{max} / z_{ref} , or at the reference (calibration) depth

Tap **Add**. The machine becomes active.

10:27

Cancel **New Machine** Add

MANUFACTURER
Select manufacturer...

MACHINE ID / SERIAL
e.g. H1912345

REFERENCE ABSORBED DOSE PROTOCOL
IAEA TRS 398 Rev.1 (2024)

SETUP CONFIGURATION
SSD-Based Setup

DOSE REPORTING POINT
Dose at D_{max} — based on PDD₁₀ (%)

Reference-depth dose scaled to d_{max} using PDD₁₀ per energy.

10:28

CONFIGURATION SETTINGS
Facility · Machine · Detector · Beam Libraries · Backup Management

! Lock a record to protect it · Import or export your settings

FACILITY **+ Add**

General Hospital
Radiation Oncology **ACTIVE**

+ Add Machine

BACKUP & RESTORE

Export everything — facilities, machines, detectors (with certificates), electrometers, beams (with factors such as K_{ph} , K_a , K_{phs} , etc.) and saved reports — to one file you can move to another phone, then Import it there.

Export Backup **Import Backup**

LEGAL

User Agreement & Disclaimer
Clinical-use responsibility & verification terms

RadIX Dose - Version 2.2 (4)
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Home Output Factors Results Settings

New Machine — pick the code of practice. The expanded facility with “Add Machine”.

3

Add a detector (ionization chamber).

Open the machine's **Detector library** and add a chamber:

- Pick a model from the built-in catalog (e.g. an IBA FC65-G Farmer) or define your own
- Enter its **serial number** and **cavity volume / length**
- Enter the $N_{D,w}$ calibration certificate — value, date, and validity period — and choose the ^{60}Co or **cross-calibration** route

Save. Detector cards warn you as the certificate nears expiry.

14:17

Cancel **New Detector** Add

2.31 cm

Auto-filled from the detector library (vendor brochure). Tap the pencil to override.

BEAM CALIBRATION ROUTE ($N_{D,w}$)

^{60}Co Calibration

SERIAL NUMBER

1001

☒ DOSE-TO-WATER CALIBRATION(S) — $N_{D,w}$ PER ELECTROMETER

ELECTROMETER

IBA — DOSE-1 · S/N: 2001

CALIBRATION FACTOR $N_{D,w}$ (Gy / nC)

0.04776

CALIBRATION VOLTAGE

+ - e.g. 300 V

TRACK CALIBRATION EXPIRY ☐

CALIBRATION CERTIFICATE, ISSUING LABORATORY

PSDL, SSDL or Other...

Add PDF Certificate

PDF only · File size can up to 8,4 MB

+ Add electrometer & $N_{D,w}$

New Detector — chamber, serial, and the $N_{D,w}$ calibration per electrometer.

4

Add an electrometer (optional).

In the **Electrometer library**, add make/model/serial and, if applicable, a calibration factor k_{elec} (default 1.000).

11:06

Cancel New Electrometer Add

MANUFACTURER

IBA

MODEL

DOSE-1

SERIAL NUMBER

2001

ELECTROMETER CALIBRATION FACTOR (K_{Elec})

1.000

i Under IAEA TRS-398, record $K_{Elec} = 1.000$ when the ionization chamber and electrometer are calibrated together, since the combined $N_{p,w}$ already contains the electrometer response. Enter a separate K_{Elec} from the electrometer's own calibration certificate (in nC/rdg, or a dimensionless value near unity for a charge-reading electrometer) only when the chamber and electrometer are calibrated separately.

New Electrometer — make, model, serial, and k_{elec} .

5

Add a beam.

In the **Beam library**, add a **Photon** (or Electron) beam:

- **Energy** — e.g. 6 MV
- The **beam-quality index** your protocol needs — $\text{TPR}_{20,10}$, $\%dd(10)_x$, or R_{50}
- **Calibration conditions** — field size, delivered MU, and expected dose (used for the output check), plus PDD/TMR at the reference depth

Save.

14:21

Cancel New Photon Beam Add

ENERGY (MV)

6 FFF

DOSE MAXIMUM (D_{Max}) (g/cm^2)

1.5

APPROX. D_{Max} (g/cm^3)

1.50

PERCENTAGE DEPTH DOSE at 10cm (PDD_{10}) (%)

66.5

TISSUE-PHANTOM RATIO 20cm / 10cm ($TPR_{20/10}$)

0.67

New Photon Beam — energy, PDD_{10} , and $TPR_{20,10}$.

DEMO VALUES — REPRODUCE THE WORKED EXAMPLE

Enter these to reproduce the calculation shown in the next section, then adjust any of them to explore the app.

Machine	Varian TrueBeam · ID LINAC-1 · IAEA TRS-398 (2006) · SSD-based · reference depth 10 cm · 10×10 cm · 100 MU → 100 cGy expected
Electrometer	IBA DOSE-1 · S/N 2001 · k_{elec} 1.000
Detector	IBA FC65-G Farmer · S/N 1001 · 0.65 cm ³ · cavity 2.31 cm · $N_{D,w}$ 0.04776 Gy/nC · ⁶⁰ Co route
Photon beam	6 MV · TPR_{20,10} 0.670 · PDD ₁₀ 66.5% · d_{max} 1.5 cm
Conditions & reading	20.0 °C · 1013.3 hPa → k_{TP} 1.000 · M = 14.02 nC (single reading)
Correction factors	k_Q 0.9957 · k_{pol} 1.0010 (M_+ 10, M_- 10.02) · k_s 1.0030 (M_1 10.0000, M_2 9.9390, V_1/V_2 3, Pulsed) · k_{pw} 1.0000 · k_{elec} 1.0000
Result	≈ 100.6 cGy at d_{max} (100.648) · +0.65 % vs 100 cGy expected

Note: the app requires each detector and beam to be **locked** (Settings ▶ Detector / Beam Library) before it computes a dose, and the $k_{\text{pol}}/k_s/k_{\text{pw}}$ factors are set on the **Factors** tab. Any value can be changed at any time.

⚡ Shortcut — load a ready-made example

To skip the manual setup, use **Settings → Import Backup** and select a sample backup file: it loads a complete demo clinic (facility, machine, chamber, beam, and saved records) in one step. Export Backup does the reverse — everything to one shareable file.

03 Run a calculation

Open the **Output** tab. Your active facility, machine, detector and beam are shown at the top.

1 Check the conditions.

Confirm **temperature** and **pressure** — the air-density correction k_{TP} is derived automatically.

2 Select the detector and beam.

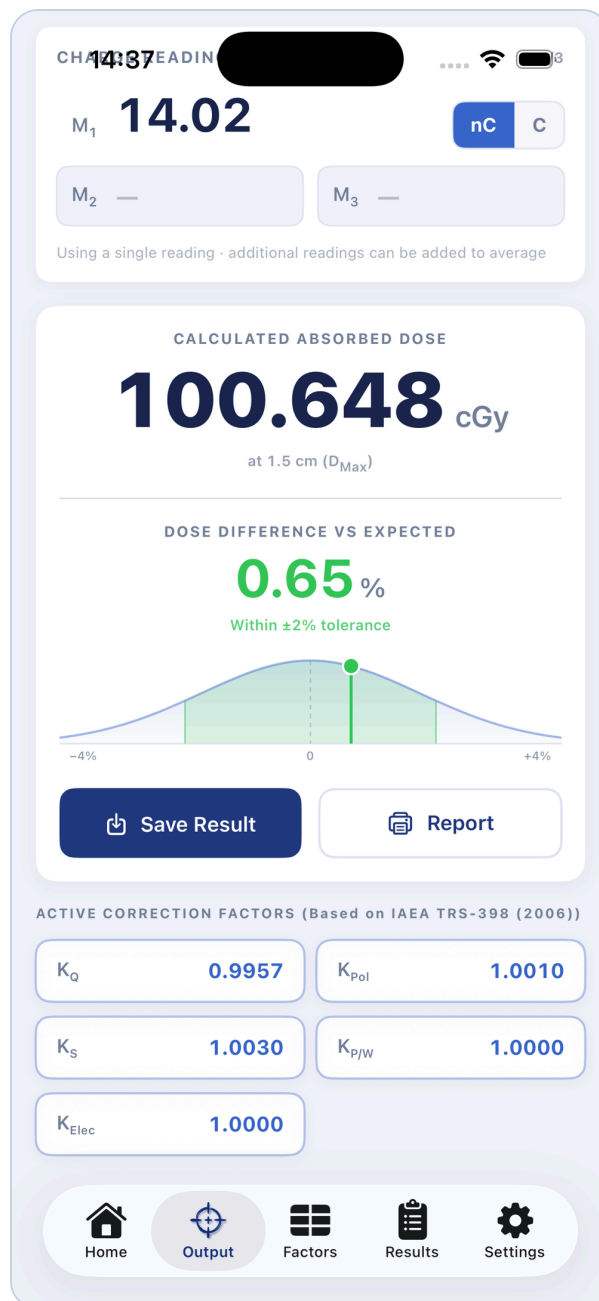
Tap your chamber under *Detector Information*, and your energy under *Beam Information* (Photon / Electron).

3 Enter the charge reading.

Type M_1 (add M_2/M_3 to average up to three). The **absorbed dose to water** and the **% deviation vs expected** appear live on the gauge.

4 Save or report.

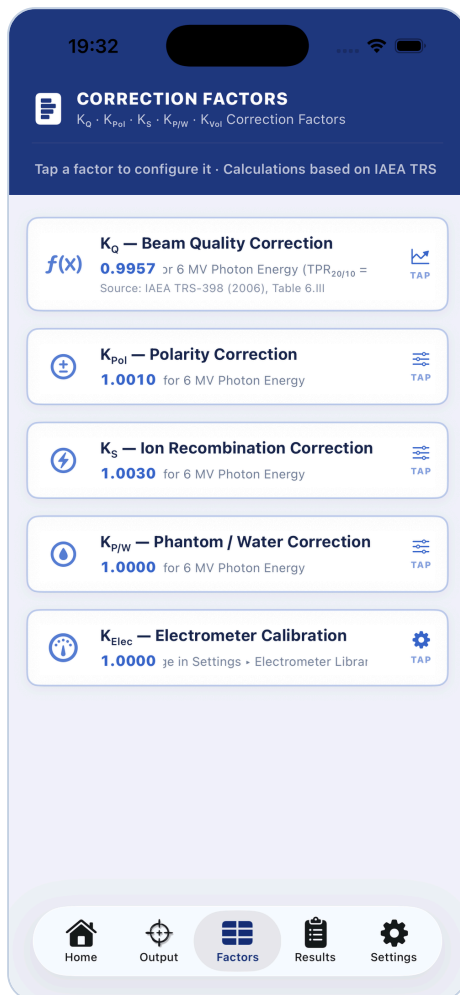
Tap **Save Result** to record it, or **Report** to generate a PDF.



Output — the chamber reading becomes an absorbed dose (here 100.648 cGy, +0.65 % within tolerance), with the active correction factors listed below.

04 Correction factors

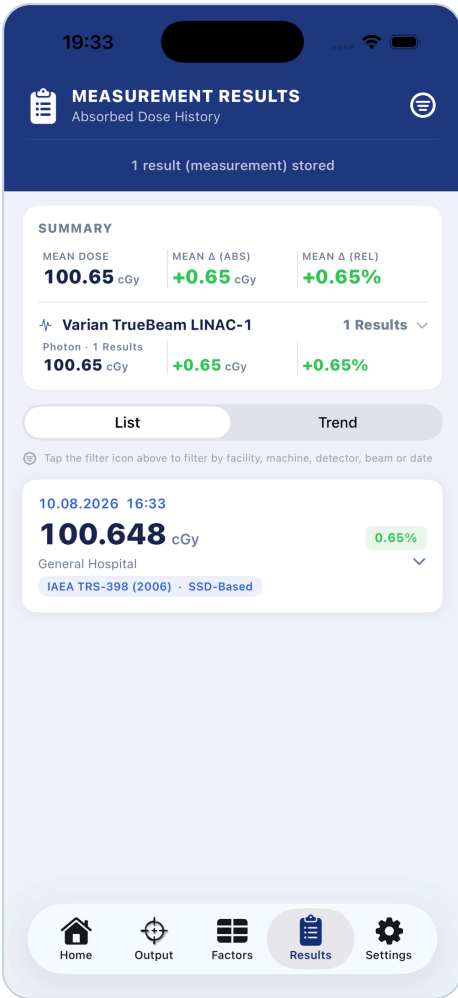
Open the **Factors** tab to measure and lock the corrections that aren't derived live — K_{pol} (polarity), K_S (recombination), $K_{P/W}$ (phantom/water), and P_{gr} for TG-51 electrons. Each is stored **per chamber and energy**. A measured factor stays out of the dose until you **lock** it, so a half-entered value never silently changes a result.



Factors — every correction is shown and traceable; tap one to configure it.

05 Records & reporting

The **Results** tab keeps every saved calculation with its full factor snapshot — filterable by date and machine, with a scrolling trend chart so drift is visible at a glance. Use **Settings** → **Export Backup** to preserve or move your whole database.



Results — the summary and every saved record; switch to Trend for the drift chart.

To generate a report, tap **Report** in the **Output** view (next to *Save Result*). Tick the records to include, then **Preview & Export**.

Close

Reporting Records

✓

○

○

≡

✓

Aug 10, 2026 · 6 MV

100.65 cGy

Varian TrueBeam LINAC-1 · FC65-G Far...

+0.65%

1 of 1 selected

AP

Preview & Export

Reporting Records — select the measurements to include, then Preview & Export.

The result is a signed-off **A4 landscape PDF** — a trend chart above a full factor table (M , k_{TP} , k_Q , k_{Pol} , k_S , $k_{P/W}$, k_{Vol}), dose, deviation, and a Pass/Fail per record, with *Performed by* / *Reviewed by* sign-off lines — shareable or printable from the export button.

Reference Dosimetry Report

Tolerance $\pm 2\%$

All measurements

Generated Aug 10, 2026 at 17:03 · 1 measurement · Page 1 of 1

+2%

+1%

+0%

-1%

-2%

Aug 8

Aug 9

Aug 10

Aug 11

● TrueBeam LINAC-16 MV

Date	Machine	Detector	Beam / Field	M (nC)	K_{TP}	K_Q	K_{Pol}	K_S	$K_{P/W}$	K_{Vol}	Dose (cGy)	$\Delta\%$	Result
Aug 10, 2026	TrueBeam LINAC-1	FC65-G Farmer... $N_{D,w}$: 0.04776	6 MV 10 x 10 cm	14.0200	1.0000	0.9957	1.0010	1.0030	1.0000	N/A	100.65 1.5 cm	+0.65	PASS

Performed by

Reviewed by

IAEA TRS 398 (2006) - absorbed dose to water - this report does not replace institutional QA procedures.

A4 · Landscape — the exported Reference Dosimetry Report (chart + full factor table, Pass/Fail per record).

Remember

RadiX Dose supports professionals — it never replaces the protocols, your judgement, or independent verification. Always verify every input, factor, and dose against the relevant code of practice before clinical use.

RadiX Dose v2.3 · Getting Started. A hands-on companion to the app — build an example clinic, then run your first reference-dosimetry calculation. © 2026 Oğuzhan Ayrancıoğlu (IAEA TRS-398 · AAPM TG-51).